CS482/682 Final Project Report 11

Monocular Depth-Estimation

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1 Introduction

Background Traditional supervised depth estimation models require large datasets with depth ground truth labels. Collecting such labeled data is often difficult and time-consuming. To approach this problem, we use unsupervised learning to train a deep neural network on rectified RGB videos and use the model to make monocular depth estimations.

Related Work Zhou et. al [1] approached this problem with an unsupervised learning framework based on monocular video sequences. Given a target single-view image, they propose a conjunctive Depth (DispNet)/Pose CNN to output the depth map. The DispNet comprises of an encoder, consisting of convolutional layers and a decoder, with upsampling and concatenation layers for a series of four separate predictions. It also employs skip connections between corresponding same-size layers, akin to U-net.

Chen et. al [2] had another approach using SEBottleNeck, consisting of a Bottleneck Block followed by a *Squeeze-and-Excitation* (SE) block. The motivation for SE-blocks are its capacity to learn dependencies between certain channels to assign weights to each channel, shifting the attention of the model to what matters. It does this by pooling each channel into a single scalar value for a $1 \times 1 \times C$ vector, followed by two linear layers to get the weights. It then goes back to the original input and multiplies each channel by its corresponding weight. *Parametric Rectified Linear Unit* (PReLU) is used in SEBlock to prevent neurons from dying out to gradients stuck at 0, while still being computationally efficient.

2 Methods

Taking inspirations from these previous approaches, we wanted to implement SEBottleNeck into DispNet to assess their improvements in the context of depth prediction. We also wanted to evaluate how the DispNet architecture may perform under other modifications, namely colored data augmentation and reduced training data size.[1]

Setup Dataset: We used the KITTI dataset, and for the sake of computational expenses, we resorted to using only 3GB and 94MB of the original training data size. For our implementation, we followed the standard Eigen split [2] and resized the images to 416×128. The images went through data augmentations and were then organized into batches, with the stereo pairs normalized before being fed into our network.

Training cycles: For training all of our models we used 200 epochs. We used the Adam optimizer with $\beta_1 = 0.9$ and $\beta_2 = 0.999$. Our learning rate for all models was 0.0002. We implemented data augmentation for all models, which was horizontal flip with probability 0.5 and zooming in on the image by 0-15% and then cropping it. We also implemented Xavier Uniform initialization for weights of all convolutional layers. To ensure the reproducibility of our results, we set a random seed for all models.

Models Base Model: The original model from Zhou et. al [1] trained on 3GB data.

[1] Our code will be made available at `https://github.com/ivan-fan-dk/DL-final-project`

[2] The Eigen split is a split previously used for the KITTI dataset and is unremarkable aside from the specific numbers of images put into each category.[3]

SE: This version incorporates SEBottleneck modules in the encoder of DispNet, to dynamically recalibrate feature maps, enhancing channel-specific relevance. We also included PReLU over the previous ReLU function. The reduction parameter for SE-blocks was kept at 16 from Hu's original paper [4], because of its reported balance of speed and efficacy.

Aug: This is the base model but trained with more types of data augmentation, specifically color, for robustness to different visual anomalies. These include altering the gamma of an entire batch by up to ±20%, altering the brightness of an entire batch by a factor of 0.5-2, and color shifts, which is the brightness transformation applied to each color channel independently with a factor of 0.8-1.2.

Light: This is the base model but trained only on 94MB of data to observe the impact that the size of the training dataset has for performance.

3 Evaluation & Results

Metrics: We used four different error scores: Scale-Invariant Log Error (SILog), Relative Squared Error (sqErrorRel), Absolute Squared Error (absErrorRel), and Root Mean Squared Error (RMSE).

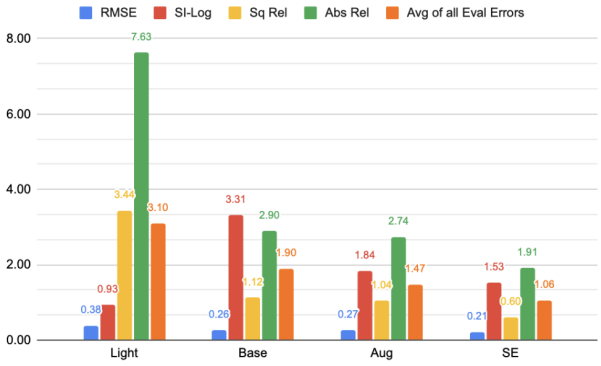

Figure 1: Bar Graph of Evaluation Errors per Model.

Figure 1 shows the results of our four models from six test images w.r.t the average of the respective error metrics. In the Apppendix, each of the four models has results from six test images. Each sample consists of the test image used, the model's prediction, and an image representing the model's error. We also generated histograms for depth predictions and the normalized ground truth values to give insight on whether the predictions are aligned with the true distribution.

4 Discussion

Our experimental results aligned with expectations while revealing interesting model behaviors. The SE model demonstrated superior performance across all metrics, with its RMSE of 0.21 representing a 20% improvement over the base model's 0.26. The substantial improvements in squared relative error (0.60 vs base's 1.12) and absolute relative error (1.91 vs 2.90) suggest that SE-blocks are particularly effective at reducing large prediction errors.

The histogram analysis of depth predictions revealed distinctive behaviors across models. While Base, Aug, and SE showed prediction distributions closely matching ground truth, the Light model's predictions clustered around 0.5 on the normalized depth scale. This conservative behavior with only 94MB of training data explains its higher error metrics (RMSE: 0.38, Sq Rel: 3.44) but interestingly resulted in the best SI-Log score of 0.93, suggesting consistent relative depth relationships despite limited training.

Aug's improvements over the base model, particularly in SI-Log error (from 3.31 to 1.84), demonstrate the value of data augmentation for model robustness. However, a limitation in our evaluation stems from potential dataset bias, as images in our training dataset were captured in daytime conditions. This may create unintended correlations between brightness and depth, where the sky is typically both the brightest and furthest part of the image.

Future work could explore combining the SE model's architectural improvements with Aug's color-based data augmentations. Given their complementary strengths in accuracy and robustness, a hybrid approach could yield a more generalizable model that maintains high accuracy while better handling real-world visual variations.

References

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